ELK406E
Distribution of Electrical Energy

Prof.Dr. Mustafa BAĞRIYANIK

E-mail: bagriy@itu.edu.tr

Conductor Sizing

- To find the proper conductor size,
  - the type of conductor (copper or aluminum),
  - the distance in km,
  - the current that are needed for the load

  are must be known.
- There are an amazing variety of sizes and types of conductors.
- Several electrical, mechanical, and economic characteristics affect conductor selection

Ampacity — The peak current-carrying capability of a conductor limits the current (and power) carrying capability.

Economics — Often we will use a conductor that normally operates well below its ampacity rating.

- The cost of the extra aluminum pays for itself with lower I^2R losses; the conductor runs cooler.
- This also leaves room for expansion.

- ***Mechanical strength*** — Especially on rural lines with long span lengths, mechanical strength plays an important role in size and type of conductor.
  - Stronger conductors like ACSR are used more often.
  - Ice and wind loadings must be considered.

- ***Corrosion*** — While not usually a problem, corrosion sometimes limits certain types of conductors in certain applications

CHAPTER SIX

Electric Lines

Overhead Lines

Article 42- The following provisions apply for all outdoors overhead electric lines within the scope of the regulation.

Conductors and insulators

Article 43- a) Bare conductors:

1) Characteristics and use of the conductors:

i) The conductors must be made of copper, solid aluminum, steel stranded aluminum or other alloys which are equivalent to those in terms of strength and chemical durability. Conductors must comply with the relevant standards.

ii) Single wire (solid) or braided steel conductors may only be used if they are coated with a metal cover so as to endure the corrosion effects that may occur at their place of use continuously.

iii) Aluminum conductors, whatever their cross section and type, and copper conductors having cross sections greater than 16 mm^2 (including 16 mm^2) used in overhead lines must be braided.

vii) Cross-sections of the bare braided conductors used in overhead lines may not be less than the following values.

	LV	**HV**
Copper	10 mm^2	16 mm^2
Solid aluminum	21 mm^2	--
Steel / aluminum	--	21/4 mm^2
Steel	16 mm^2	16 mm^2
Bronze	16 mm^2	16 mm^2

Single wire or braided copper conductors with 10 mm^2 cross sections or other conductors equivalent in terms of conductivity may be used on low voltage lines at short distances.

Ampacity

- ***Ampacity*** — The peak current-carrying capability of a conductor limits the current (and power) carrying capability

The thermal rating of conductor is dependent on the following factors :

A. Meteorological/Environmental Conditions
    A. Solar radiation.
    B. Wind velocity.
    C. Ambient temperature.

B. Conductor surface Characteristics
    A. Age of the Conductor.
    B. Maximum design conductor temperature.
    C. Allowable losses of strength of aluminum metal at design temperature.

Kod	Tel Sayısı ve Tel Çapı / Number and Wire Diameter		Dış Çap	Kesit Alanı / Cross Section			Net Ağırlık (Yaklaşık)	Kopma Yükü	D.A. Direnci	Akım Taşıma Kapasitesi
Code	Alüminyum / Aluminium	Çelik / Steel	Overall Diameter	Alüminyum / Aluminium	Çelik / Steel	Toplam / Total	Net Weight (Approx)	Rated Stenght	DC Resistance	Cuurent Carrying Capacity
-	mm	mm	mm	mm	mm	mm	kg / km	kN	ohm/km	A
Wren	6/1.33	1/1.33	3.99	8.39	1.42	9.81	34	3.3	3.4226	63
Warbler	6/1.50	1/1.50	4.5	10.59	1.34	11.93	43	4.2	2.7139	67
Turkey	6/1.68	1/1.68	5.04	13.29	2.19	15.48	54	5.2	2.1535	86
Thrush	6/1.89	1/1.89	5.67	16.77	2.77	19.54	68	6.5	1.7077	93
Swan	6/2.12	1/2.12	6.36	21.16	3.55	24.71	85	8.2	1.3537	109
Swallow	6/2.38	1/2.38	7.14	26.65	4.45	31.09	108	10.0	1.0738	126
Sparrow	6/2.67	1/2.67	8.01	33.61	5.61	39.22	136	12.4	0.8504	140
Robin	6/3.00	1/3.00	9	42.39	7.1	49.49	171	15.5	0.6752	162
Raven	6/3.37	1/3.37	10.11	53.48	8.9	62.38	215	19.0	0.5351	186
Quail	6/3.78	1/3.78	11.34	67.42	11.23	78.65	273	10.8	0.4245	211
Pigeon	6/4.25	1/4.25	12.75	85.03	14.19	99.22	343	29.7	0.3366	241
Penguin	6/4.77	1/4.77	14.31	107.23	17.87	125.1	433	37.5	0.2671	276
Owl	6/5.36	7/1.74	16.09	135.16	17.55	152.7	508	42.5	0.2119	322
Waxwing	18/3.09	1/3.09	15.15	135.16	7.48	142.6	430	31.5	0.2126	319
Partridge	26/2.57	7/2.00	16.28	135.16	22	157.2	545	50.01	0.2136	321

LOSSES:

- Line losses are from the line current flowing through the resistance of the conductors.
- After distribution transformer losses, primary line losses are the largest cause of losses on the distribution system.
- Like any resistive losses, line losses are a function of the current squared multiplied by the resistance (I^2R).

Losses

The real power losses due to the load currents in the conductors of the single-phase two-wire ungrounded lateral with full-capacity neutral is

$$P_{Losses,1ph} = I^2 (2R) \quad W$$

The real power losses due to the load currents in the conductors of the equivalent three-phase four-wire balanced lateral is

$$P_{Losses,3ph} = 3\, I^2\, R \quad W$$

VOLTAGE REGULATION

- Voltage drop is higher with lower voltage distribution systems, poor power factor, single-phase circuits, and unbalanced circuits.
- Distribution utilities have several ways to control steady-state voltage.

The most popular regulation methods to improve and control the voltage include

1. Use of Load Tap Changing (LTC) transformers.
2. Application of voltage regulators in the distribution substation as well as on the feeders.
3. Application of shunt (or series) capacitor on the feeders or at the distribution substation. Fixed and switched capacitors

4. Balancing of loads on primary feeders.
5. Increasing of feeder conductor size.
6. Increasing or upgrading primary voltage level.
7. Changing feeder sections from single-phase to three-phase.
8. Transferring of loads from existing feeders to new feeders.
9. Installation of new substations and primary feeders.

While these options are usually considered during the planning stages, LTCs, regulators, and capacitors provide the best means of achieving good voltage regulation during the operational stages on a continuous basis.

- Most utilities use LTCs to regulate the substation bus and supplementary feeder regulators and/or switched capacitor banks where needed.
- Taps on distribution transformers are another tool to provide proper voltage to customers.
- Distribution transformers are available with and without no-load taps (meaning the taps are to be changed without load)

- The standards specify the acceptable operational ranges at two locations on electric power systems.
    - Service *voltage is the point where the electrical systems of the supplier and the user are interconnected*. (normally at the meter) Maintaining acceptable voltage at the service entrance is the utility's responsibility
    - Utilization voltage *is the voltage at the line terminals of utilization equipment*. This voltage is the facility's responsibility.
- Equipment manufacturers should design equipment which operates satisfactorily within the given limits

Load Types

The modeling and analysis of a power system depend upon the load.

- Spot Load (lumped load)

They are considered as known loads with metered demands that have constant values.

- Distributed Loads

They represent customer loads spread along section.

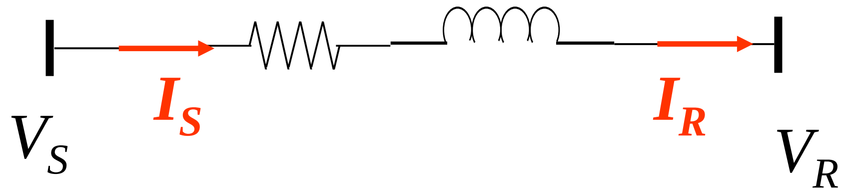
Voltage Drop

- A voltage drop in an electrical circuit normally occurs when current is passed through the wire
- greater the resistance of the circuit, the higher the voltage drop
- a voltage drop of 5% at the furthest receptacle in a branch wiring circuit is acceptable for normal efficiency
- In a 240 volt circuit, this means that there should be no more than a 12 volt drop (228 volts) at the furthest point.

- Voltage drop shall not be below 5% at the system.
- To find the proper conductor size,
  - the type of conductor (copper or aluminum),
  - the distance in km,
  - the current that are needed for the load

  are must be known.

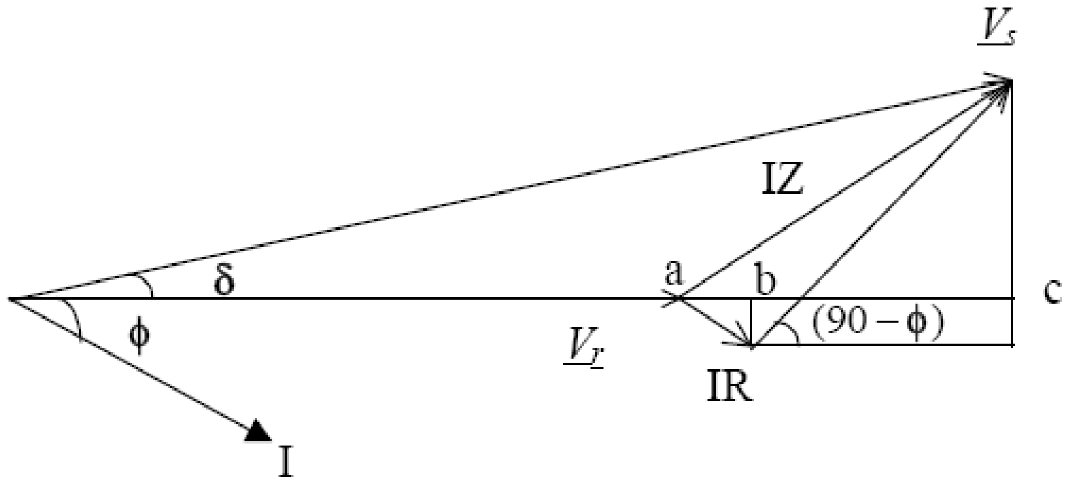
$$Z = R + jX = |Z|\angle\theta^{\circ}$$



- Z is the total impedance of the line,

$$Z = z.l \quad [\Omega]$$

- Here, *z* is the impedance of the per-unit length of the line and, *l* is the length of the feeder main [km].



- Taking the receiving end voltage as the reference phasor

$$V_R = V_r \angle 0°$$

- and, the sending end voltage is

$$V_S = V_s \angle \delta$$

The Voltage Drop (VD) is

$$\Delta V = Z.I$$

The percent Voltage Regulation is defined as

$$\%VR = \frac{V_s - V_r}{V_r}.100$$

If the complex power of the load that connected at the receiving end of the feeder is

$$S = P + jQ = V_R I^*$$

The current can be calculated as

$$I = \frac{S^*}{V_R} = \frac{P - jQ}{V_R}$$

The sending end voltage is

$$V_S = V_R + Z I$$

$$V_S (Cos\delta + j\,Sin\delta) = V_r \angle 0° + I\left(Cos\phi + j\,Sin\phi\right)\left(R + j\,X\right)$$

Approximate Method for Voltage Regulation:

In typical distribution systems

$$R \cong X$$

and the voltage angle δ is closer to zero or typically

$$0 \le \delta \le 4°$$

Therefore, for a typical distribution circuit,

$Sin\delta$ can be neglected.

Hence,

$$V_S = V_s \, Cos\delta$$

and Equation $V_S \left(Cos\delta + j\, Sin\delta \right) = V_r \angle 0^\circ + I \left(Cos\phi + j\, Sin\phi \right) \left(R + j\, X \right)$

becomes (for lagging load)

$$V_S \, Cos\delta = V_r \angle 0^\circ + I \left(Cos\phi - j\, Sin\phi \right) \left(R + j\, X \right)$$

and

$$V_S \cong V_r + I\, RCos\phi + IXSin\phi$$

The Voltage Drop along the feeder is

$$VD = I\,RCos\phi + IXSin\phi$$

The percent Voltage Regulation is

$$\%VR = \frac{I\,RCos\phi + IXSin\phi}{V_r}.100$$

$$VD = I\,RCos\phi + IXSin\phi$$

- The VD is positive quantity.
- The VD is negative when there is a leading power factor due to shunt capacitors or when there is a negative reactance X due to series capacitors installed in the circuits

In terms of the load power factor, pf, the real and reactive line currents are

$$I_R = I\,pf = I\cos\phi$$

$$I_X = I\,qf = I\sin\phi$$

where

I =magnitude of the line current, A

pf = load power factor

qf = load reactive power factor = $\sin(\cos^{-1}(pf))$

ϕ□= angle between the voltage and the current

- In practice, the approximate voltage drop is a very acceptable measure since the error between the exact value and approximate value is negligible.
- The approximate voltage drop is quite accurate for most distribution situations.
- The largest error occurs under heavy current and leading power factor.

- The approximation has an error less than 1% for an angle between the sending and receiving end voltages up to 8^{o} (which is unlikely on a distribution circuit).
- Most distribution programs use the full complex phasor calculations, so the error is mainly a consideration for hand calculations.

The approximation highlights **two important aspects about voltage drop**:

Resistive load

- At high power factors, the voltage drop strongly depends on the resistance of the conductors.
- At a power factor of 0.95, the reactive power factor (*qf*) is 0.31; so even though the resistance is normally smaller than the reactance, the resistance plays a major role.

Reactive load

- At moderate to low power factors, the voltage drop depends mainly on the reactance of the conductors.
- At a power factor of 0.8, the reactive power factor is 0.6, and because the reactance is usually larger than the resistance, the reactive load causes most of the voltage drop.
- Poor power factor significantly increases voltage drop.

Exact Method for Voltage Regulation:

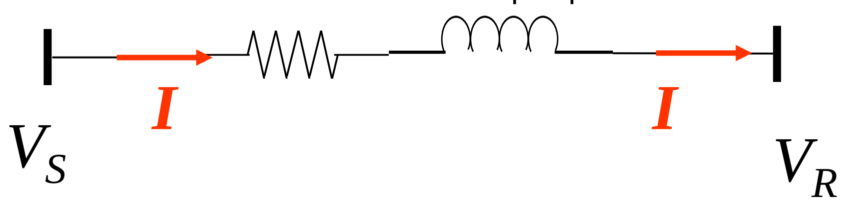
The exact value of VD can be calculated using the following equation;

$$V_S = V_R + Z\,I$$

$$V_S\,(Cos\delta + j\,Sin\delta) = V_r\angle 0^\circ + I\left(Cos\phi + j\,Sin\phi\right)\left(R + j\,X\right)$$

If V_R is known, it is easy to solve.

$$Z = R + jX = |Z|\angle\theta^\circ$$



$$\boxed{V_S(Cos\delta + j\,Sin\delta) = V_r\angle 0° + I\left(Cos\phi + j\,Sin\phi\right)\left(R + j\,X\right)}$$

- However, in distribution systems, usually $\boldsymbol{V_S}$, the voltage at the substation **is controlled** and **held constant** for varying loading (I) conditions, which imply V_R and hence the drop changes.
- Therefore, in situations where $\boldsymbol{V_S}$**,** $\boldsymbol{I}$**, and** $\boldsymbol{Z}$ **are known** then $\boldsymbol{V_R}$ **can be obtained** only by an iterative method as explained when discussing **power flow studies**.

Expanding the equation with *Vs* as reference:

$$V_s\angle 0° = V_r\angle -\delta + Z\angle\theta\, I\angle(-\delta-\varphi)$$

In this equation *Vr* and δ are unknowns. Resolving this equation into real and imaginary parts:

$$V_s = V_r\, Cos\delta + IZ\, Cos(\theta-\delta-\phi)$$
$$0 = V_r\, Sin\delta + IZ\, Sin(\theta-\delta-\phi)$$

By solving these equations simultaneously, *Vr* and δ can exactly be obtained.

- This exact solution is possible only for a point-to-point two bus example.
- For radial systems involving more than 2 buses, the problem becomes a power flow problem.
- Either the solution can be obtained for a balanced case by considering only one-phase iterative method such as Newton-Raphsom method, or in a more general sense, by three-phase iterative method such as Source-to-Load Iterative method.

- The effective impedance Z of the three-phase main depends upon the nature of the load.
- For example, when a lumped-sum load is connected at the end of the main, the effective impedance is,

$$Z = z.l \quad \left[\Omega / phase\right]$$

where z is the impedance of three-phase main line [Ω/km], l is the length of the feeder main [km] example

An example for voltage regulation calculation by the approximate method.

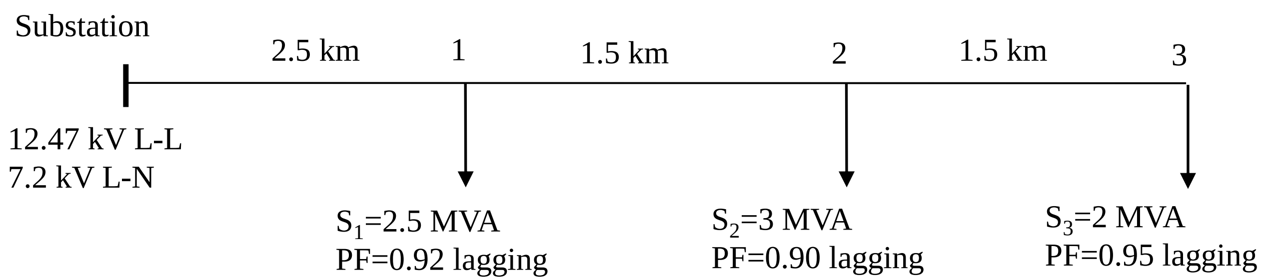
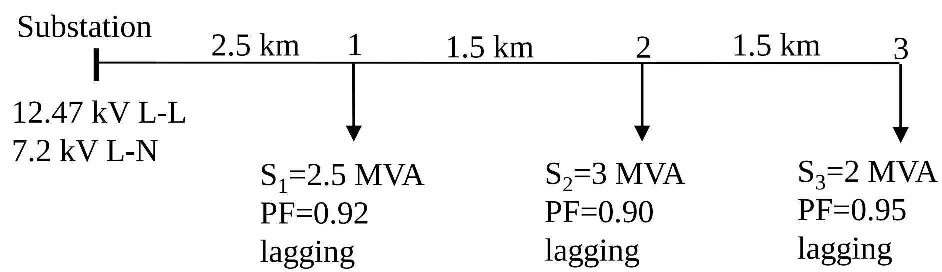


Figure: A simple feeder system supplying three point loads

- Assume that a three-phase feeder main with the series impedance $Z=0.1603 + j\,0.4127\ \Omega/km = 0.4427\ \angle 68.78\ \Omega/km$.
- Find the percent voltage drop across the entire primary feeder for the stipulated load conditions, using the approximate method.

example



Solution:

$$Zij = Z \text{ x length of feeder}$$

$$Z_{S1} = (0.4427\angle 68.78°)\ \Omega/\text{km x } 2.5\text{ km} = 1.10675\angle 68.78°\ \Omega$$

$$Z_{S1} = 0.4006 + j1.03171\Omega$$

$$Z_{12} = (0.4427\angle 68.78°)\ \Omega/\text{km x } 1.5\text{ km} = 0.66405\angle 68.78°\ \Omega$$

$$Z_{12} = 0.362 + j0.619\ \Omega$$

$$Z_{23} = (0.4427\angle 68.78°)\ \Omega/\text{km x } 1.5\text{ km} = 0.66405\angle 68.78°\ \Omega$$

$$Z_{23} = 0.362 + j0.619\ \Omega$$

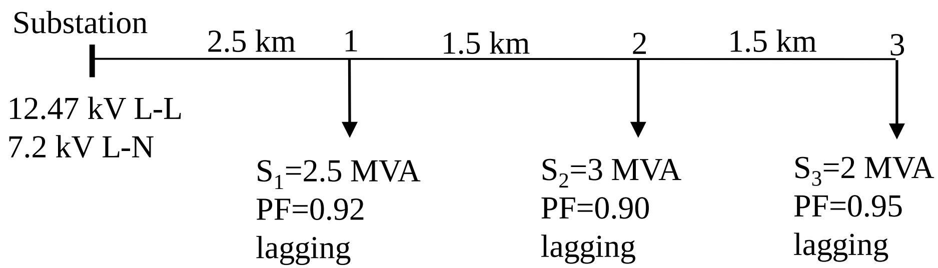
example

- To determine the approximate percent voltage drop across each section of the line, we use equation

$$VD = I\,RCos\phi + IXSin\phi$$

- In this approximate method, we will also assume that the voltage at each point remains at $7{,}200\angle 0°$ on a line-to neutral basis and then determine I in each section.

example



The current drawn by load 3 is

$$I_{L3} = \left(\frac{S_3}{V_3}\right)^* = \frac{2.0\,\mathrm{x}10^6}{\sqrt{3}\,\mathrm{x}12.47\mathrm{x}10^3}\angle -(\cos^{-1} 0.95)$$

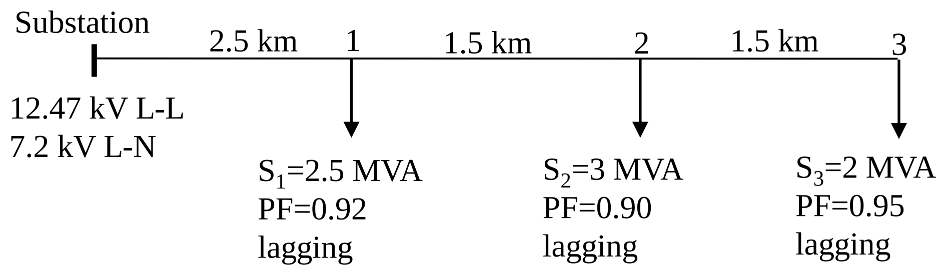
$$I_{L3} = 92.60\angle -18.19°\ \mathrm{A}$$

Similarly, the current drawn by load 2 is

$$I_{L2} = \left(\frac{S_2}{V_2}\right)^* = \frac{3.0\,\mathrm{x}10^6}{\sqrt{3}\,\mathrm{x}12.47\mathrm{x}10^3}\angle -(\cos^{-1} 0.90)$$

$$I_{L2} = 138.89\angle -25.84°\ \mathrm{A}$$

example



The current drawn by load 1 is

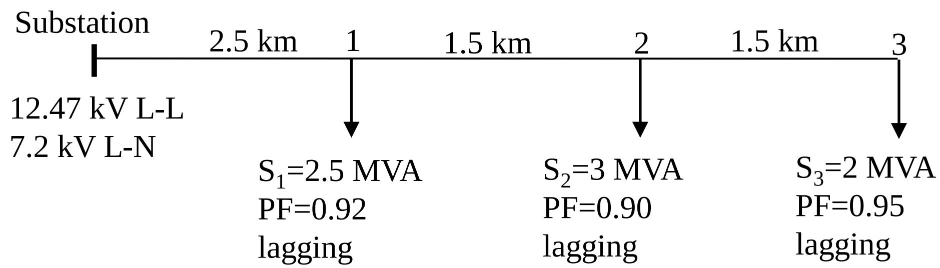
$$I_{L1} = \left(\frac{S_1}{V_1}\right)^* = \frac{2.5\,\mathrm{x}10^6}{\sqrt{3}\,\mathrm{x}12.47\mathrm{x}10^3}\angle -(\cos^{-1} 0.92)$$

$$I_{L1} = 115.74\angle -23.07^\circ \text{ A}$$

The current flowing between buses 2 and 3 can be calculated as

$$I_{23} = I_{L3} = 92.60\angle -18.19^\circ \text{ A}$$

example



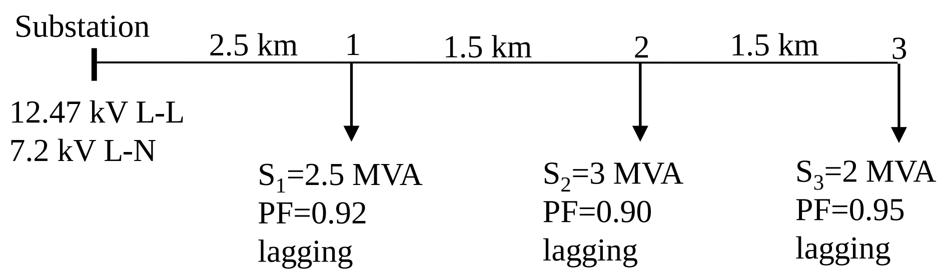
The current flowing between buses 1 and 2 can be calculated as

$$I_{12} = I_{L2} + I_{23} = 138.89\angle -25.84° \text{ A} + 92.60\angle -18.19° \text{ A}$$
$$I_{12} = 212.98 - j89.44 = 230.995\angle -22.78° \text{ A}$$

The current flowing between Substation and Bus 1 can be calculated as

$$I_{S1} = I_{L1} + I_{12} = 115.74\angle -23.07° \text{ A} + 230.995\angle -22.78° \text{ A}$$
$$I_{S1} = 319.46 - j134.8 = 346.73\angle -22.88° \text{ A}$$

example



The voltage drop between Buses 2 and 3 is

$$VD_{23} = I_{23}R_{23}\cos(18.19°) + I_{23}X_{23}\sin(18.19°)$$

$$VD_{23} = 92.6 \text{x} 0.362 \text{x} \cos(18.19°) + 92.6 \text{x} 0.619 \text{x} \sin(18.19°)$$

$$VD_{23} = 31.85 + 17.89 = 49.74 \text{ V}$$

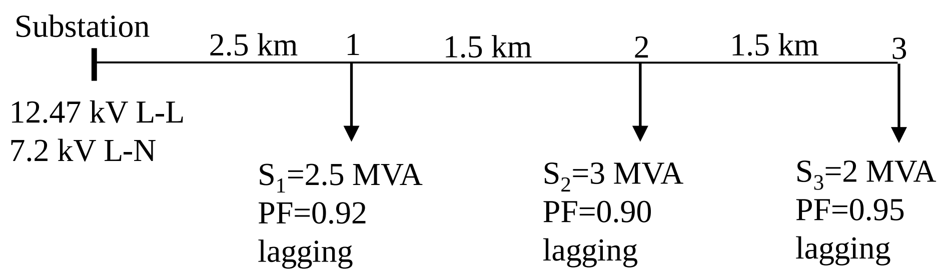
The voltage drop between Buses 1 and 2 is

$$VD_{12} = I_{12}R_{12}\cos(18.19°) + I_{12}X_{12}\sin(18.19°)$$

$$VD_{12} = 230.995 \text{x} 0.362 \text{x} \cos(22.78°) + 230.995 \text{x} 0.619 \text{x} \sin(22.78°)$$

$$VD_{12} = 77.098 + 55.36 = 132.46 \text{ V}$$

example



The voltage drop between Buses S and 1 is

$$VD_{S1} = I_{S1}R_{S1}\cos(18.19°) + I_{S1}X_{S1}\sin(18.19°)$$

$$VD_{S1} = 346.73\text{x}0.4006\text{x}\cos(22.88°) + 346.73\text{x}1.03171\text{x}\sin(22.88°)$$

$$VD_{S1} = 127.97 + 139.08 = 267.05\ \text{V}$$

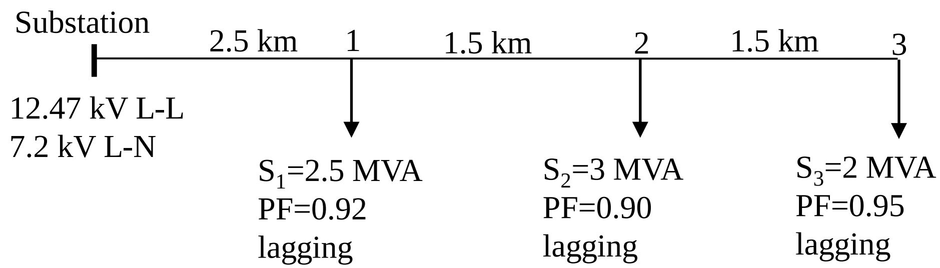
Total voltage drop in the entire feeder = sum of voltage drops

$$\sum VD = 49.74 + 132.46 + 267.05 = 449.25\ \text{V}$$

The value as a percentage of system voltage is

$$\frac{449.25}{7200}\text{x}\,100 = 6.24\ \%$$

example



EXACT METHOD

The given example can be solved by exact method.

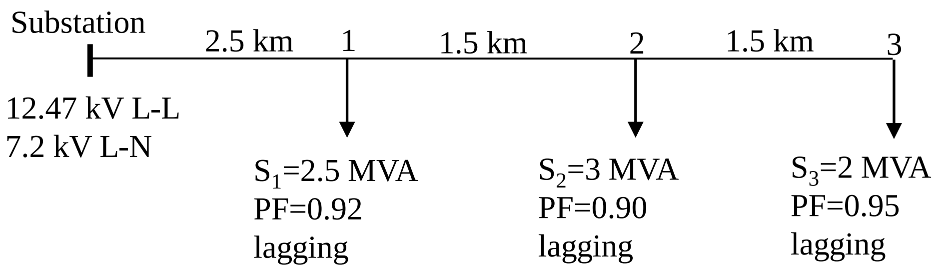
We assume that $V_3 = \dfrac{12.47}{\sqrt{3}} = 7200\angle 0^\circ \text{ V}$

$$V_2 = V_3 + I_{23} Z_{23}$$

$$V_2 = 7200\angle 0^\circ + 92.60\angle -18.19^\circ \text{ x } (0.362 + j\,0.619)$$

$$V_2 = 7249.7 + j\,43.991 \text{ V} = 7249.99\angle 0.3477^\circ \text{ V}$$

example



$$I_{L2} = 138.89\angle(0.3477 - 25.84)^\circ \text{ A} = 138.89\angle -25.4923^\circ \text{ A}$$

$$I_{12} = I_{L2} + I_{23} = 138.89\angle -25.4923^\circ \text{ A} + 92.60\angle -18.19^\circ \text{ A}$$

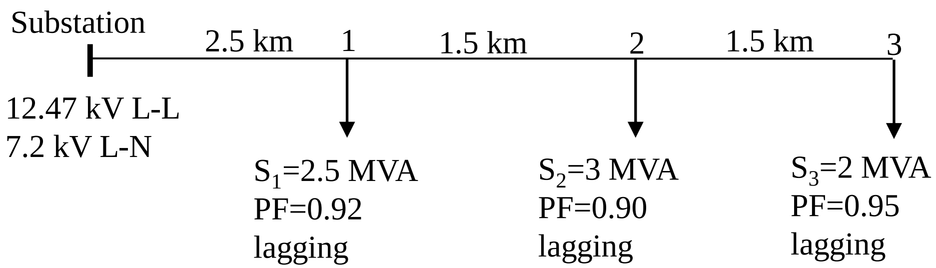
$$I_{12} = 213.34 - j88.684 = 231.0389\angle -22.5722^\circ \text{ A}$$

$$V_1 = V_2 + I_{12} Z_{12}$$

$$V_2 = 7249.99\angle 0.3477^\circ + 231.0389\angle -22.5722^\circ \text{ x } (0.362 + j0.619)$$

$$V_2 = 7382.0 + j143.95 \text{ V} = 7383.4\angle 1.1171^\circ \text{ V}$$

example



$$I_{L1} = 115.74\angle(1.1171 - 23.07)^\circ \text{ A} = 115.74\angle -21.9529 \quad \text{A}$$

$$I_{S1} = I_{L1} + I_{12} = 115.74\angle -21.9529^\circ \text{ A} + 230.995\angle -22.78^\circ \text{ A}$$

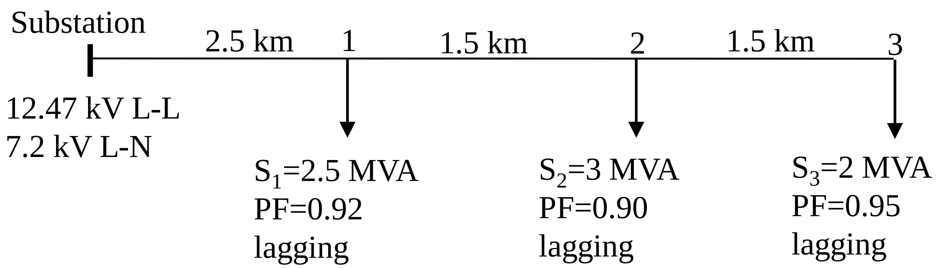
$$I_{S1} = 320.32 - j132.71 = 346.727\angle -22.5039^\circ \text{ A}$$

$$V_S = V_1 + I_{S1}\, Z_{S1}$$

$$V_2 = 7383.4\angle 1.1171^\circ + 346.727\angle -22.5039^\circ \text{ x } (0.4006 + j1.03171)$$

$$V_2 = 7647.2 + j421.27 \quad \text{V} = 7658.8\angle 3.1531^\circ \quad \text{V}$$

example

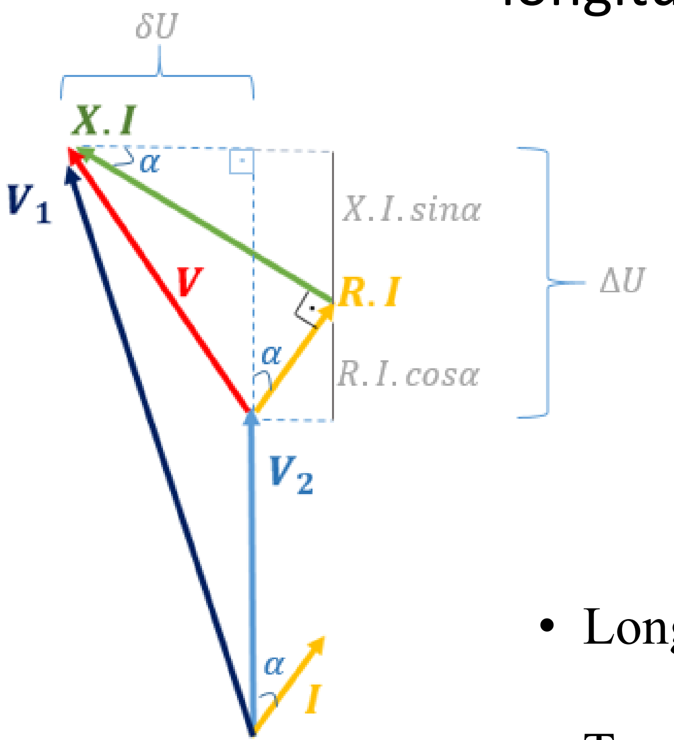


The percentage Voltage Regulation is

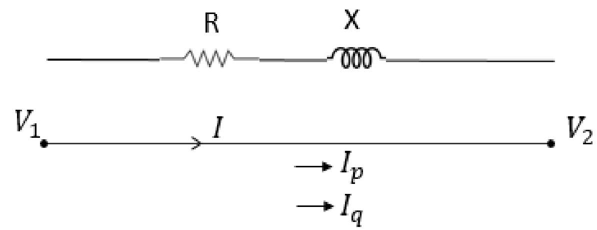
$$\%VR = \frac{V_s - V_r}{V_r} \text{x}100 = \frac{7658.8 - 7200}{7200} \text{x}100 = \frac{458.8}{7200} \text{x}100 = 6.37\%$$

- It can be seen from the results that there are no significant difference.
- The approximate method involves less computation and, it is therefore adopted.

Voltage Drop



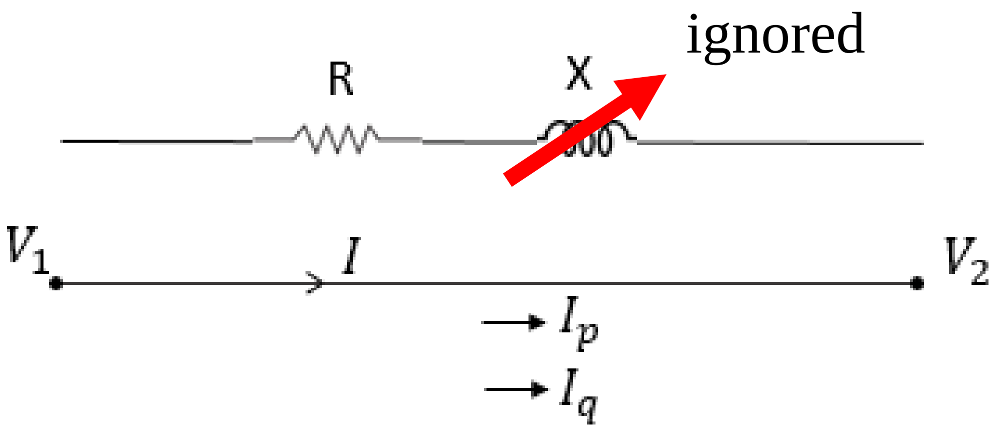
- Voltage drop has two components, a longitudinal and a transversal component



- Longitudinal $Voltage\ Drop,\ \Delta U = R\ I_p + X\ I_q$
- Transversal $Voltage\ Drop,\ \delta U = X\ I_p - R\ I_q$

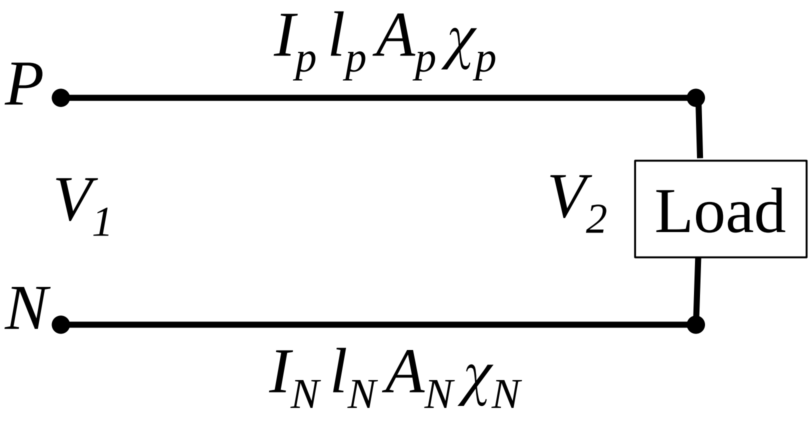
Calculating VD in LV distribution systems

- If A<16mm2,
  X is neglected



$$Voltage\ Drop, \qquad \Delta U = R\ I_p = R\ I\ Cos\varphi$$

Single phase radial system

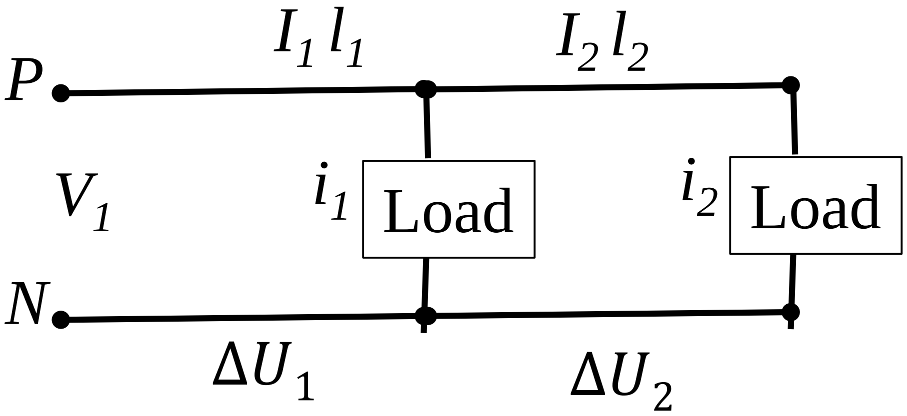


$$\Delta U = \Delta UP + \Delta UN = RP\ I_p + R_N\ I_N = \frac{l_P}{x_p A_p}\ I_p + \frac{l_N}{x_N A_N}\ I_N$$

If same conductors are used for both phase and neutral connection

$$\Delta U = 2\ \Delta UP = 2\ R_P\ I_p = 2\frac{l_P}{x_p A_p}\ I_p$$

Existing multiple-customers in single phase two wire network



$$I_2 = i_2$$
$$I_1 = i_1 + i_2$$

$$\Delta U = \Delta U_1 + \Delta U_2 = 2\, R_1\, I_1 + 2\, R_2\, I_2 = 2\,\frac{l_1}{x_1 A_1}\, I_1 + 2\,\frac{l_2}{x_2 A_2}\, I_2$$

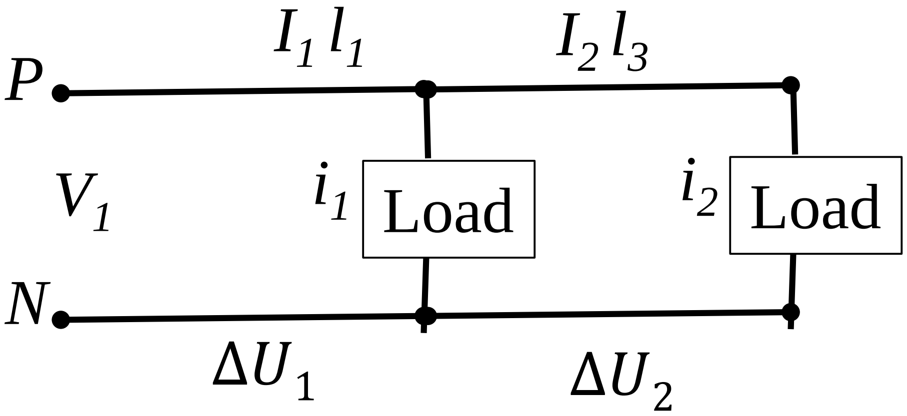
If $x = x_1 = x_2$ $\qquad A = A_1 = A_2$

If same conductors are used for both phase and neutral connection

$$\Delta U = \frac{2}{x\, A}\, [I_1\, l_1 + I_2\, l_2]$$

using line currents

Existing multiple-customers in single phase two wire network



$$L_1 = l_1$$
$$L_2 = l_1 + l_2$$

$$\Delta U = \Delta U_1 + \Delta U_2 = 2\,R_1\,I_1 + 2\,R_2\,I_2 = 2\frac{l_1}{x_1 A_1}\,I_1 + 2\frac{l_2}{x_2 A_2}\,I_2$$

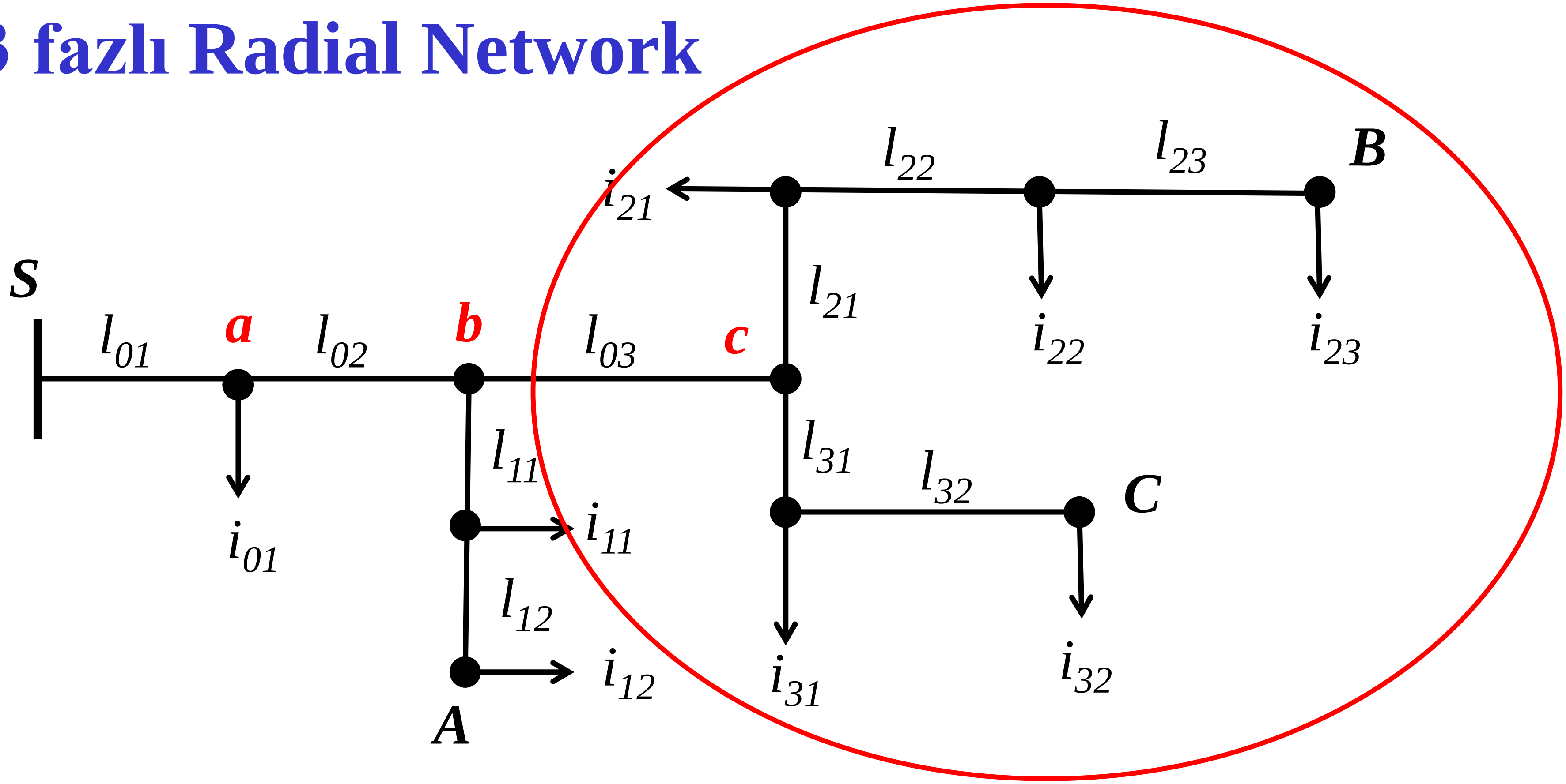
If $x = x_1 = x_2$ $\qquad A = A_1 = A_2$

If same conductors are used for both phase and neutral connection

$$\Delta U = \frac{2}{x\,A}\,[i_1 L_1 + i_2 L_2]$$

using load currents

3 fazlı Radial Network



$$\Delta U = \frac{1}{x\,A}\,K\,[i_{01}\,l_{01} + (l_{01} + l_{02})\,i_b + i_{11}\,(l_{01} + l_{02} + l_{11}) + i_{12}\,(l_{01} + l_{02} + l_{11} + l_{12})]$$

$$K = (1 + \frac{X}{R}\tan\varphi)$$

Radial Feeders with Uniformly Distributed Load

- We can assume that there are many closely spaced loads and/or lateral lines connected to the main.
- Since the load is uniformly distributed along the main, the load current in the main is a function of the distance.
- Approximations using uniform load distributions are useful.

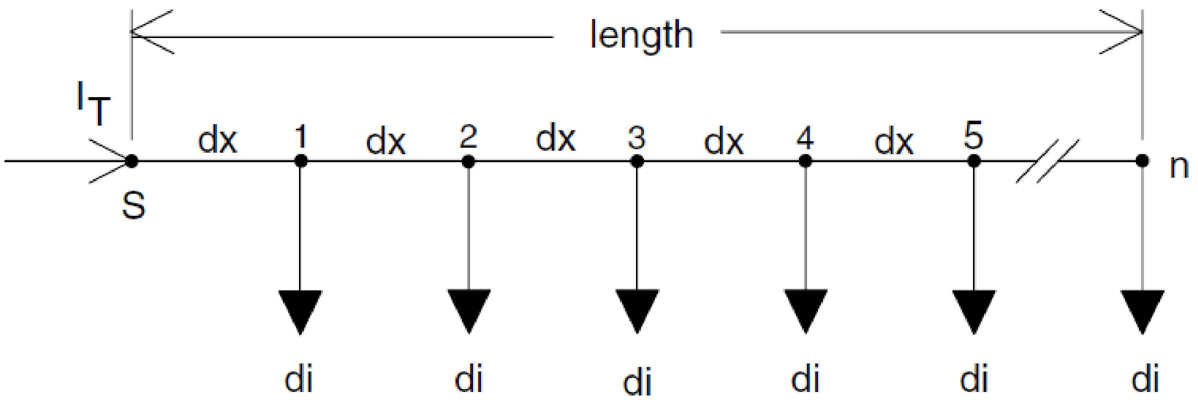
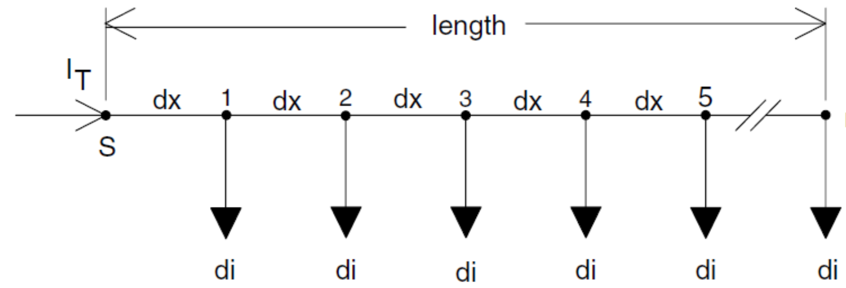


Figure shows a generalized line with n uniformly distributed loads

The loads are all equal and will be treated as constant current loads with a value of *di*

The total current into the feeder is I_T.



It is desired to determine the total voltage drop from the source node (S) to the last node n.

l = length of the feeder

$z = r + jx$ = impedance of the line (in ohm/km)

dx = length of each line section

di = load currents at each node

n = number of nodes and number of line sections

I_T = total current into the feeder

The load currents are given by:

$$di = \frac{I_T}{n}$$

The voltage drop in the first line segment is given by:

$$Vdrop_1 = \mathrm{Re}\{z \cdot dx \cdot (n \cdot di)\}$$

The voltage drop in the second line segment is given by:

$$Vdrop_2 = \mathrm{Re}\{z \cdot dx \cdot [(n-1) \cdot di]\}$$

The total voltage drop from the source node to the last node is then given by:

$$Vdrop_{\text{total}} = Vdrop_1 + Vdrop_2 + \cdots + Vdrop_n$$
$$Vdrop_{\text{total}} = \mathrm{Re}\{z \cdot dx \cdot di \cdot [n + (n-1) + (n-2) + \cdots + (1)]\}$$

Equation can be reduced by recognizing the series expansion

$$1 + 2 + 3 + \cdots + n = \frac{n(n+1)}{2}$$

Using the expansion, Equation becomes:

$$Vdrop_{\text{total}} = \text{Re}\left\{z \cdot dx \cdot di \cdot \left[\frac{n \cdot (n+1)}{2}\right]\right\}$$

The incremental distance is

$$dx = \frac{l}{n}$$

The incremental current is

$$di = \frac{I_T}{n}$$

Substituting dx and di into equation-Vdrop

$$Vdrop_{\text{total}} = \text{Re}\left\{z \cdot \frac{l}{n} \cdot \frac{I_T}{n} \cdot \left[\frac{n \cdot (n+1)}{2}\right]\right\}$$

$$Vdrop_{\text{total}} = \text{Re}\left\{z \cdot l \cdot I_T \cdot \frac{1}{2} \cdot \left(\frac{n+1}{n}\right)\right\}$$

$$Vdrop_{\text{total}} = \text{Re}\left\{\frac{1}{2} \cdot Z \cdot I_T \cdot \left(1 + \frac{1}{n}\right)\right\}$$

where $Z = z \cdot l$

- Last equation gives the general equation for computing the total voltage drop from the source to the last node *n* for a line of length *l*.
- In the limiting case where *n* goes to infinity, the final equation becomes:

$$Vdrop_{\text{total}} = \text{Re}\left\{\frac{1}{2} \cdot Z \cdot I_T\right\}$$

- In the equation, *Z* represents the total impedance from the source to the
- end of the line.
- The voltage drop is the total from the source to the end of the line.
- The equation can be interpreted in two ways.
- The first is to recognize that the total line distributed load can be lumped at the midpoint of the lateral.
- A second interpretation is to lump one-half of the total line load at the end of the line (node *n*).

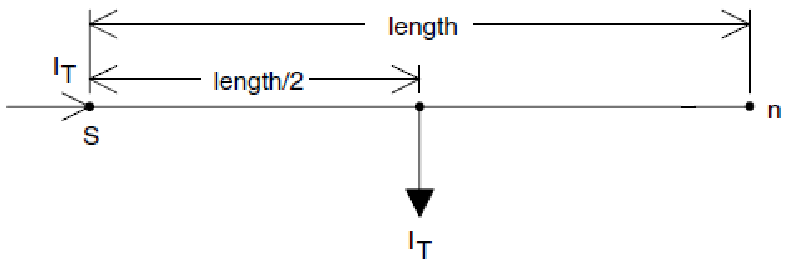


FIGURE 3.7
Load lumped at the midpoint.

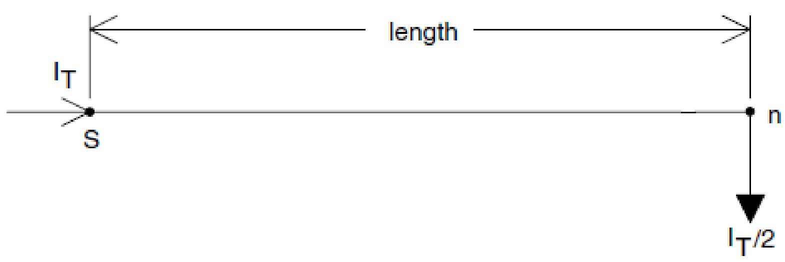
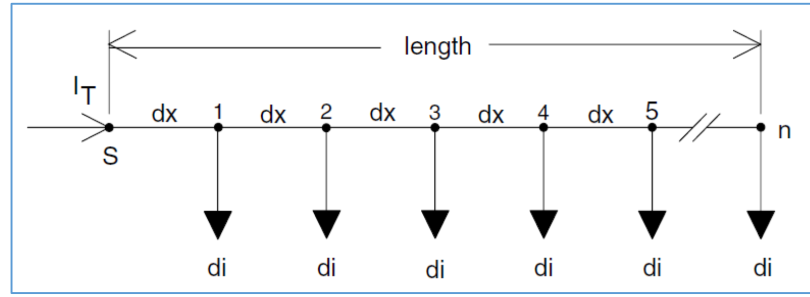


FIGURE 3.8
One-half load lumped at the end.

$$Ploss_1 = 3 \cdot (r \cdot dx) \cdot |(n \cdot di)|^2 \tag{3.20}$$

The power loss in the second segment is given by:

$$Ploss_2 = 3 \cdot (r \cdot dx) \cdot [|(n-1) \cdot di|]^2 \tag{3.21}$$



The total power loss over the length of the line is then given by:

$$Ploss_{\text{total}} = 3 \cdot (r \cdot dx) \cdot |di|^2[n^2 + (n-1)^2 + (n-2)^2 + \cdots + 1^2] \tag{3.22}$$

The series inside the brackets of Equation 3.22 is the sum of the squares of n numbers and is equal to:

$$1^2 + 2^2 + 3^2 + \cdots + n^2 = \frac{n \cdot (n+1) \cdot (2n+1)}{6} \tag{3.23}$$

Substituting Equations 3.14, 3.15, and 3.23 into Equation 3.22 gives:

$$Ploss_{\text{total}} = 3 \cdot \left(r \cdot \frac{l}{n}\right) \cdot \left(\left|\frac{I_T}{n}\right|\right)^2 \cdot \left[\frac{n \cdot (n+1) \cdot (2n+1)}{6}\right] \tag{3.24}$$

Simplifying Equation 3.24:

$$\begin{aligned}
Ploss_{\text{total}} &= 3 \cdot R \cdot |I_T|^2 \cdot \left[\frac{(n+1) \cdot (2n+1)}{6 \cdot n^2}\right] \\
Ploss_{\text{total}} &= 3 \cdot R \cdot |I_T|^2 \cdot \left[\frac{2 \cdot n^2 + 3 \cdot n + 1}{6 \cdot n^2}\right] \\
Ploss_{\text{total}} &= 3 \cdot R \cdot |I_T|^2 \cdot \left[\frac{1}{3} + \frac{1}{2 \cdot n} + \frac{1}{6 \cdot n^2}\right]
\end{aligned} \tag{3.25}$$

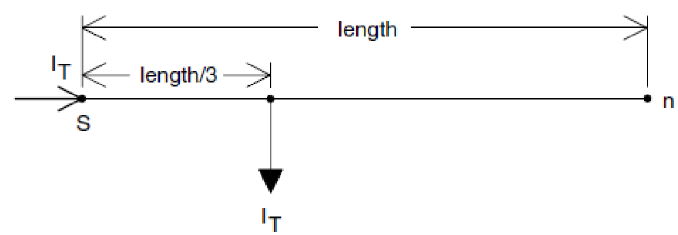
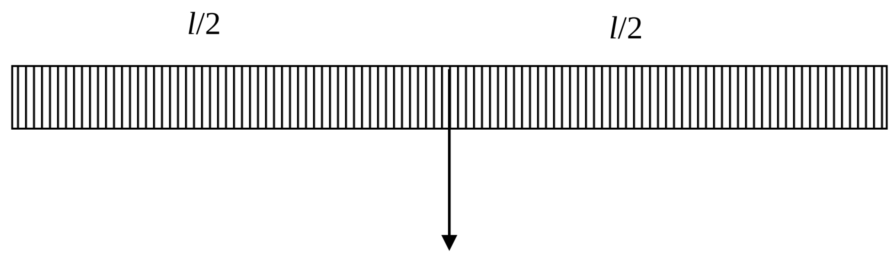


FIGURE 3.9
Power loss model.

Where $R = r \cdot l$, the total resistance per phase of the line segment, Equation 3.25 gives the total three-phase power loss for a discrete number of nodes and line segments. For a truly uniformly distributed load, the number of nodes goes to infinity. When that limiting case is taken in Equation 3.25, the final equation for computing the total three-phase power loss down the line is given by:

$$Ploss_{\text{total}} = 3 \cdot \left[\frac{1}{3} \cdot R \cdot |I_T|^2\right] \tag{3.26}$$

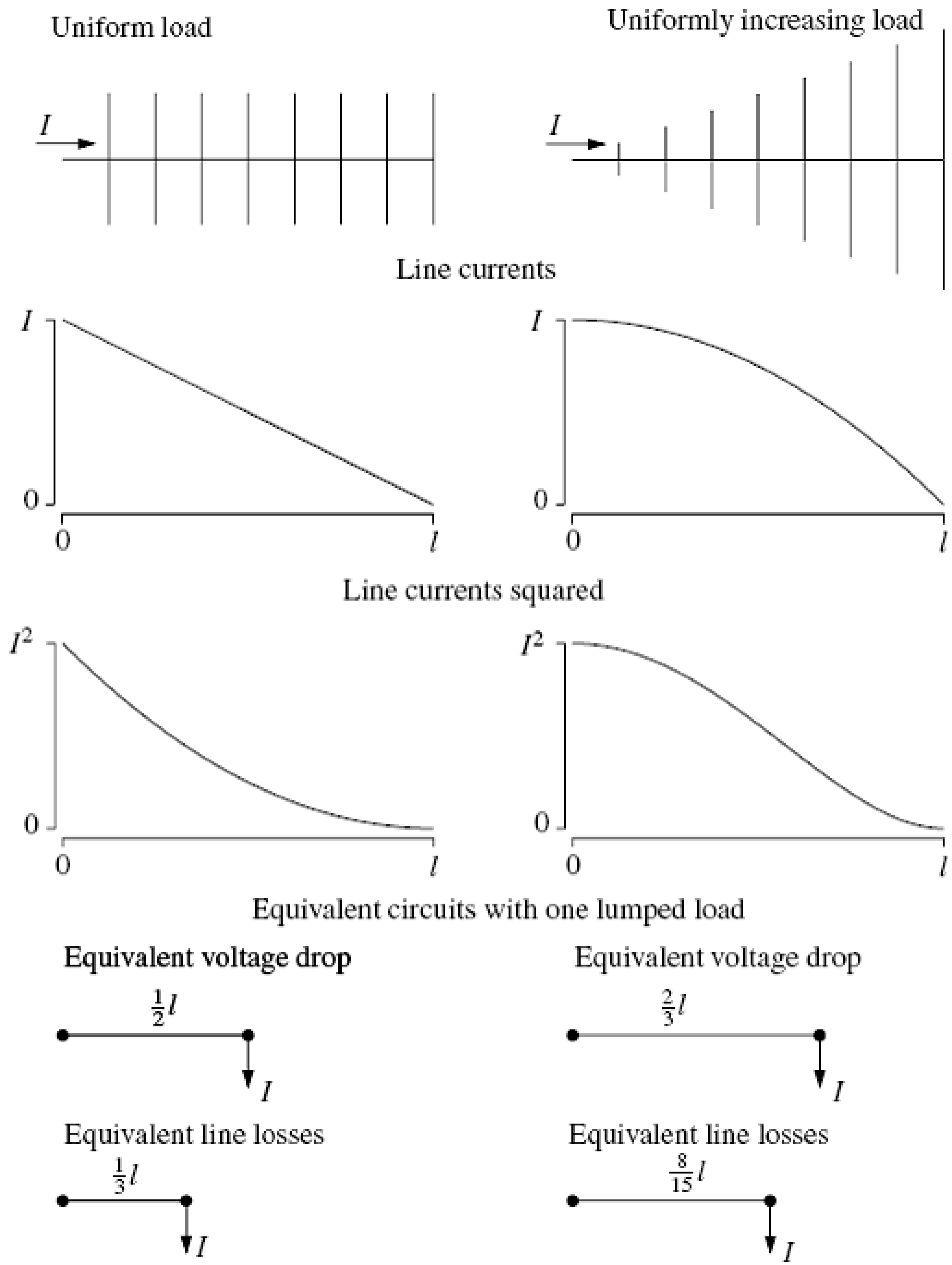
- A uniformly distributed load along a circuit of length *l* has the same losses as a single lumped load placed at a length of l/3 from the source end.
- For voltage drop, the equivalent circuits are different: a uniformly distributed load along a circuit of length *l* has the same voltage drop as a single lumped load placed at a length of l/2 from the source end.



- This 1/2 rule for voltage drop and the 1/3 rule for losses are helpful approximations when doing hand calculations or when making simplifications to enter in a load-flow program.

- If a three-phase feeder main has the tapped-off load increasing linearly with the distance (a uniformly increasing load), the equivalent lumped load is at 2/3 of the length from the source for voltage drop and, 8/15 of the length from the source for losses.

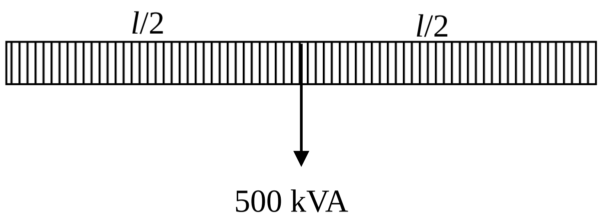
Equivalent circuits of uniform loads [Electric Power Distribution Handbook]



example

Example[T. Gönen**]:**

Assume that a three-phase 4.16 kV wye-grounded feeder main has copper conductors. The series impedance of the conductor is Z=1.503 + j 0.7456 Ω/mi. The 500 kVA (*a lagging-load power factor of 0.9*) load is uniformly distributed along the feeder main. Calculate the percent voltage drop in the main. The length of the feeder main is 1 mi. (1 mi = 1.609344 kilometers**)**



example

The effective feeder length for uniformly distributed load is

$$\frac{l}{2} = \frac{1}{2} = 0.5\,\mathrm{mi}$$

The current is

$$I = \frac{500\,kVA}{\sqrt{3}\,4.16\,kV} = 69.39\,\mathrm{A}$$

example

The Voltage Drop along the feeder is

$$VD = I\,RCos\phi + IXSin\phi$$

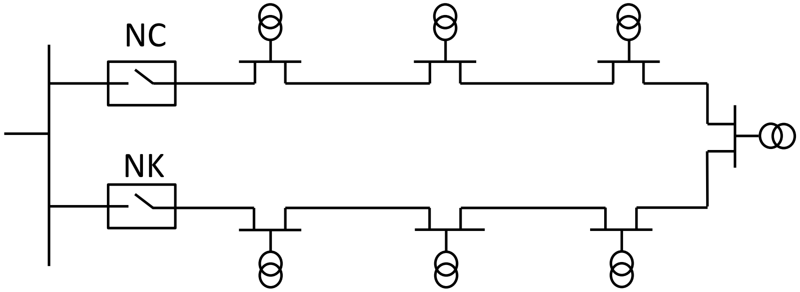
$$= 69.39\text{ A }1.503\,\Omega/\text{mi }0.5\text{ mi }0.9 + 69.39\text{ A }0.7456\,\Omega/\text{mi }0.5\text{ mi }0.4359$$

$$= 58.208\,\text{V}$$

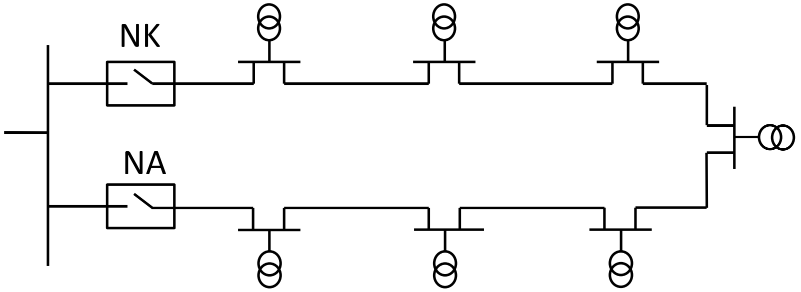
The value as a percentage of system voltage is

$$\frac{58.208}{4.16/\sqrt{3}}\,\text{x}\,100 = \frac{58.208}{2400}\,\text{x}\,100 = 2.43\,\%$$

Closed Loop



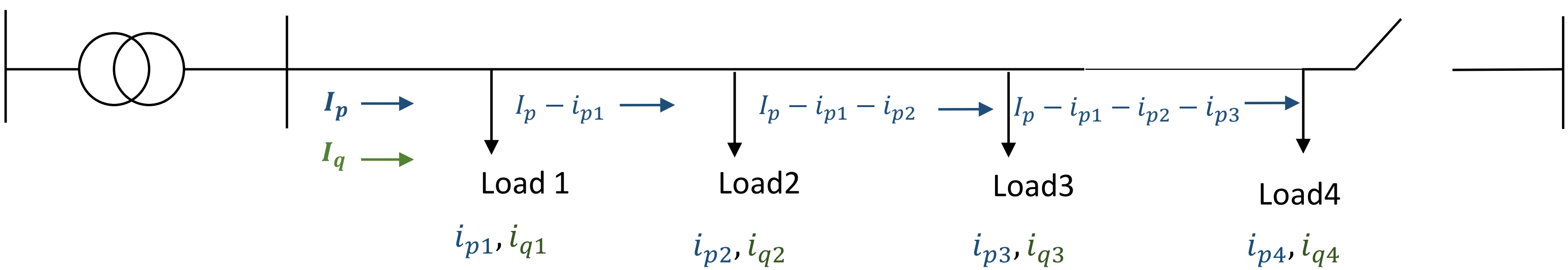
Open Loop



NC: Normaly closed

NO: Normaly open

VD (Open Loop network)



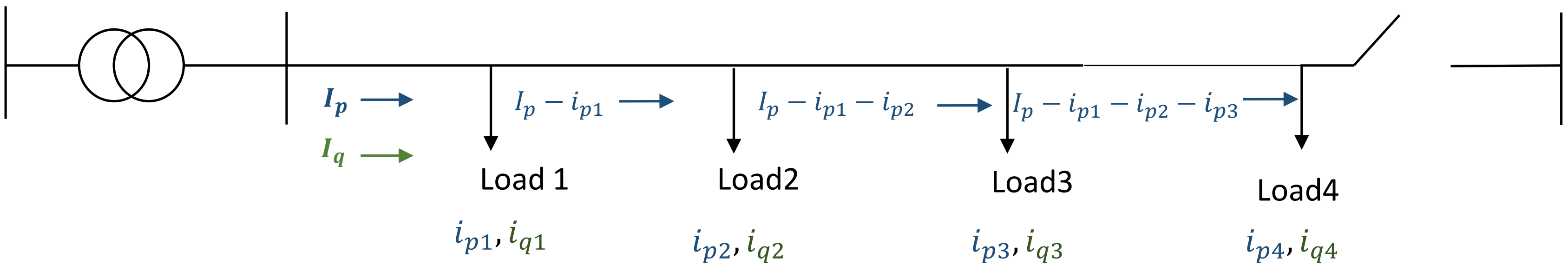
- $longitudinal$ VD, $\Delta U = R\,I_p + X\,I_q$
- $transversal\ VD,\ \delta U = X\,I_p - R\,I_q$

$$\sqrt{\Delta U^2 + \delta U^2}$$

- $R = \frac{l}{Ꝁ.q}$, l: $Line\ length[m]$; Ꝁ: $conductivity\ \left[\frac{m}{\Omega} mm^2\right]$; q: $cross\ section\ [mm^2]$

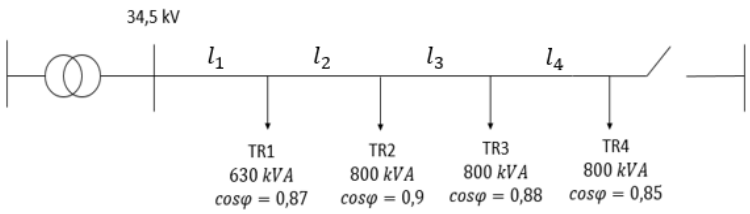
- $X = X'\,l$, $X'\ [\frac{\Omega}{\text{km}}]$; l [km]

Losses (Open Loop)



- $P_K = 3 \times R \times I^2$
- $R = \frac{l}{\text{Ꝅ}.q}$
- $P_K = \frac{3}{\text{Ꝅ}.q} \times \sum l_k \times (i_p^2 + i_q^2)$

Örnek



a) Calculate VD.

b) Calculate total Losses.

$Ꝅ = 44{,}4\ m/\Omega mm^2$

$q = 35\ mm^2$

L= $0{,}51\ mH/km$

$l_1 = l_2 = l_3 = l_4 = 650\ m$

a)

$$\%\ \varepsilon = \frac{\Delta U}{Un/\sqrt{3}}$$

$$\%\ 5 > \frac{\Delta U_{max}}{34500/\sqrt{3}}$$

$$\Delta U_{max} = 996\ V$$

$$\Delta U = R\ I_p + X\ I_q$$

$$\delta U = X\ I_p - R\ I_q$$

$$R = \frac{l}{Ꝅ.\ q}$$

$$X = X'\ l$$

TR1:

$$i_{p1} = \frac{630 \times 0{,}87}{\sqrt{3} \times 34{,}5}$$
$$= 9{,}17\ A$$

$$i_{q1} = i_{p1} \times tan\varphi$$
$$= 5{,}19\ A$$

TR2:

$$i_{p2} = 12{,}05\ A$$

$$i_{q2} = 5{,}84\ A$$

TR3:

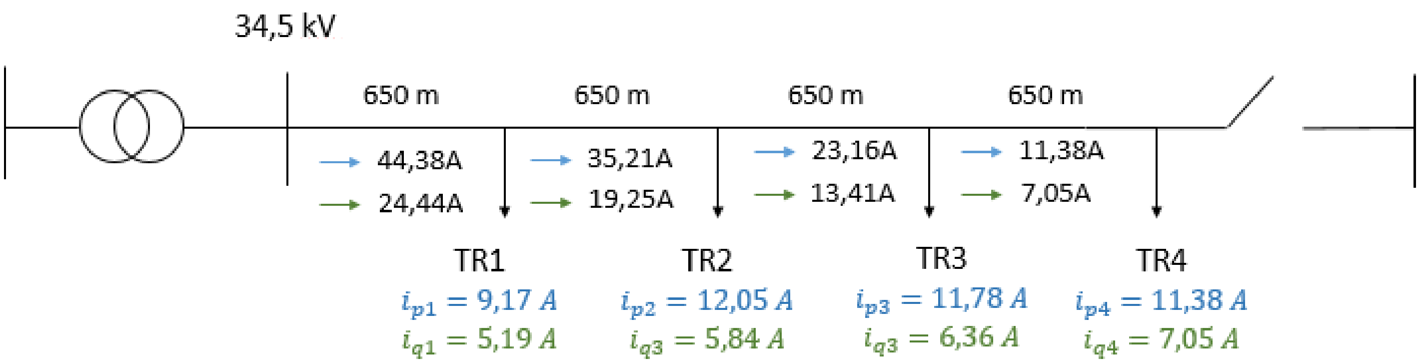
$$i_{p3} = 11{,}78\ A$$

$$i_{q3} = 6{,}36\ A$$

TR4:

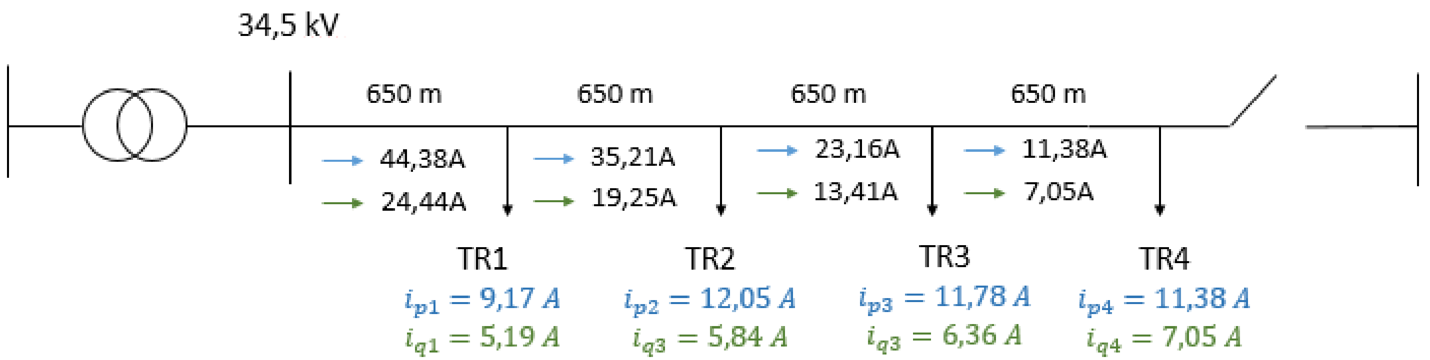
$$i_{p4} = 11{,}38\ A$$

$$i_{q4} = 7{,}05\ A$$



- $I_p = i_{p1} + i_{p2} + i_{p3} + i_{p4}$
- $I_p = 44{,}38\ A$
- $I_q = 24{,}44\ A$
- $\Delta U = R\ I_p + X\ I_q$
- $L = 0{,}51 \frac{mH}{km}$
- $X' = \omega \times L = 2\pi 50 \times \frac{0{,}51}{1000} = 0{,}16\ \Omega/\text{km}$
- $\text{X} = X' \times l$
- $\Delta U = \frac{650}{44{,}4 \times 35}\ (43{,}38 + 35{,}21 + 23{,}16 + 11{,}38) + (0{,}16 \times 0{,}65) \times (24{,}44 + 19{,}25 + 13{,}41 + 7{,}05) = 54\ V$

$$\boxed{\begin{gathered} \%\ \varepsilon = \frac{54}{34500/\sqrt{3}}\,100 \\ \%\ \varepsilon = \%0{,}3 \end{gathered}}$$



- $\delta U = X\, I_p - R\, I_q$
- $\delta U = (0{,}16 \times 0{,}65) \times (43{,}38 + 35{,}21 + 23{,}6 + 11{,}38) - \frac{650}{44{,}4 \times 35}\,(24{,}44 + 19{,}25 + 13{,}41 + 7{,}05) = -15\ V$

b) $P_K = \frac{3}{\text{K}.q} \times \sum l_k \times (i_p^2 + i_q^2)$

$$P_K = \frac{3 \times 650}{44{,}4 \times 35} \times [(44{,}38^2 + 24{,}44^2) + (35{,}21^2 + 19{,}25^2) + (23{,}16^2 + 13{,}41^2) + (11{,}38^2 + 7{,}05^2)$$

$$P_K = 6{,}37\ kW$$

$$P_{top} = 630 \times 0{,}87 + 800 \times 0{,}9 + 800 \times 0{,}88 + 800 \times 0{,}85 = 2652\ kW$$

$$\%z = \frac{P_K}{P_K + P} = \frac{6{,}37}{6{,}37 + 2652} 100 = \%0{,}24\ < \%\mathbf{8}$$

REFERENCES

- T. Gönen “Electric Power Distriburion System Engineering”
- Venkata, Lecture Notes
- Electric Power Distribution Handbook